

Homework 2

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Introduction

This project aimed to develop churn prediction models for a streaming service using various data types. Through data preprocessing and model building, I constructed a logistic regression and gradient boosting tree model. Evaluation on training and testing sets showcased their performance in terms of accuracy, recall, precision, and ROC AUC. Following model selection, I used the logistic regression model to predict churn probabilities for new customers, identifying 200 high-risk individuals. Using k nearest neighbors, I recommended content for these high-risk customers based on similar users' preferences. Ultimately, these models offer actionable insights to the streaming service, allowing someone to develop retention strategies and potentially leading to improved customer retention and revenue.

Methods

I used two distinct models for churn prediction: logistic regression and gradient boosting tree. Logistic regression predicts the likelihood of churn by fitting a logistic curve to the data, offering interpretable coefficients to understand the influence of predictor variables. Meanwhile, gradient boosting trees creates a collection of decision trees sequentially, capturing complex relationships to provide predictions. First, I conducted data preprocessing steps. I started by removing incomplete rows and standardized continuous variables like age, income, and mean hours watched using Z-score normalization to ensure uniform scales. Categorical variables such as gender, plan, and genre preferences were one-hot encoded to be able to be used in the models. I split the dataset into 90% training and 10% testing sets to evaluate model performance. Preprocessing tasks and model fitting aimed to develop accurate churn prediction models using demographic and behavioral data, which are very needed for informing retention strategies for the streaming service.

Results

Both the logistic regression and gradient boosting tree models exhibit similar performance across various metrics, indicating moderate to good predictive capabilities for identifying

potential churners in the streaming service's customer base. Despite logistic regression and gradient boosting tree showing similar model performance values, logistic regression is notably better in terms of computational efficiency, with a runtime of approximately 1 seconds, compared to gradient boosting tree, which took around 40 seconds to run. Since both models demonstrate similar accuracy, precision, recall, and ROC AUC scores on both training and testing sets, the models appear to be well fit. Additionally the high values of Accuracy, Precision and ROC AUC, and low values of Recall, as provided below, indicates a pretty good model. Considering factors such as interpretability, computational resources, and deployment requirements, the choice between logistic regression and gradient boosting tree depends on specific business needs. Logistic regression offers simplicity and interpretability but may lack the capacity to capture complex interactions, while gradient boosting tree provides flexibility and potentially higher predictive performance but requires more computational resources and careful tuning to avoid overfitting. In this case, the similar model performance in both made logistic regression a more reasonable choice considering the difference in computational efficiency. Integrating the selected classification model into the streaming service's customer management system allows for identification of high-risk customers and use of targeted retention strategies, such as personalized promotions or content recommendations. This can allow for enhanced user satisfaction, reduced churn rates, and overall increased business growth.

Performance Values	Logistic Regression	Gradient Boosting Tree
Train Acc	0.7407	0.743238
Train Precision	0.60477	0.62129
Train Recall	0.27753	0.26824
Train ROC AUC	0.7351	0.7397644
Test Acc	0.743766	0.744705
Test Precision	0.61000	0.621186
Test Recall	0.28100	0.268203
Test ROC AUC	0.74168	0.73994

Discussion/Reflection

Performing these analyses gave the comparable predictive performance of logistic regression and gradient boosting tree models for churn prediction in the streaming service domain. It was interesting to observe that despite logistic regression's simplicity and faster runtime, both models exhibited similar effectiveness. Moving forward, I would also experiment with adjusting parameters such as the train-test split ratio and the value of hyperparameters like the K value for KNN, to assess their impact on model performance and generalization.